

ACCRA TECHNICAL UNIVERSITY
Department of Computer Science

BCS 404: Introduction to Data Science with Python

**Exploratory Data Analysis, Statistical Analysis and Machine Learning
using a Real-World Dataset**

Academic Year: 2025/2026 — Second Semester

Lecturer: Dr. Joseph Dadzie

Dataset: Titanic — Machine Learning from Disaster (Kaggle)

Submission Deadline: Sunday, 19 July 2026

STUDENT NAME: BERNARD KWABLAH TETTEH

STUDENT INDEX: 01250768B

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1. Introduction

The sinking of the RMS Titanic on 15 April 1912 remains one of the most tragic maritime disasters in history. Of the estimated 2,224 passengers and crew aboard, more than 1,500 perished. This project applies Python-based data science techniques to analyse the Titanic passenger dataset sourced from Kaggle, with the aim of uncovering patterns in survival rates across passenger class, gender, age, and fare. The analysis encompasses data cleaning, exploratory visualisation, statistical analysis, and the development of a logistic regression machine learning model to predict passenger survival.

2. Dataset Description

The dataset used in this study is the Titanic training dataset obtained from the Kaggle competition titled Titanic: Machine Learning from Disaster. The dataset contains 891 observations and 12 variables capturing passenger demographics, ticket information, and survival status. Key variables include PassengerId, Survived (target variable), Pclass (passenger class), Name, Sex, Age, SibSp (siblings/spouses aboard), Parch (parents/children aboard), Ticket, Fare, Cabin, and Embarked (port of embarkation). The Survived column serves as the binary target variable where 0 indicates death and 1 indicates survival.

3. Methodology

The analysis was conducted using Python 3.12 within a Jupyter Lab environment on a Linux virtual machine. The following libraries were employed throughout the project: Pandas for data manipulation, NumPy for numerical operations, Matplotlib and Seaborn for data visualisation, and Scikit-Learn for machine learning model development. The workflow followed a structured pipeline beginning with data acquisition and cleaning, followed by exploratory visualisation, statistical analysis, and concluding with the development and evaluation of a logistic regression classifier.

4. Results

4.1 Data Acquisition

The dataset was successfully imported using the Pandas library. Initial inspection confirmed 891 rows and 12 columns. The dataset contains a mix of integer, float, and string data types. Key columns such as Age and Fare are stored as float64, while categorical variables such as Sex, Embarked, and Name are stored as object types.

4.2 Data Cleaning

Missing value analysis revealed three columns with incomplete data. The Age column contained 177 missing values representing 19.87% of entries and was imputed using the median value to avoid distortion from outliers. The Cabin column contained 687 missing values accounting for 77.10% of entries and was dropped entirely as imputation of such a large proportion would introduce significant noise. The Embarked column had only 2 missing values representing 0.22% and was filled using the mode. No duplicate rows were detected. Following cleaning, the dataset retained 891 rows across 11 columns.

4.3 Data Visualisation

Histogram of Passenger Ages: The histogram reveals that the majority of passengers were between the ages of 20 and 40, with the distribution peaking around age 30. The distribution follows a roughly unimodal and slightly right-skewed pattern, indicating that while most

passengers were young to middle-aged adults, a smaller proportion of older passengers were also present. A notable spike is observed near age 0, confirming that infants were aboard.

Bar Chart of Passenger Class Distribution: The bar chart illustrates that third class had the highest number of passengers, followed by first class and second class respectively. This suggests that the Titanic carried a significantly larger proportion of lower-fare passengers, likely emigrants and working-class travellers seeking passage to America.

Boxplot of Age by Passenger Class: First class passengers recorded the highest median age of 35 years, while second and third class passengers shared a median age of 28 years. Third class exhibited the most outliers, reflecting a wider spread of ages among lower fare passengers.

Scatter Plot of Age versus Fare: The scatter plot revealed no strong linear relationship between passenger age and fare paid. The majority of passengers across all age groups paid fares within the range of 0 to 100, suggesting age was not a determining factor in ticket pricing. A few notable outliers at significantly higher fare values likely correspond to first class passengers in premium cabins.

Correlation Heatmap: The heatmap revealed that the strongest negative correlation exists between Pclass and Fare at -0.55, confirming that passengers in higher class numbers paid lower fares. A moderate positive correlation of 0.41 was observed between SibSp and Parch, suggesting that family groups tended to travel together.

Pairplot of Selected Numerical Variables: The pairplot revealed clear differences between survivors and non-survivors. Passengers who paid higher fares were more likely to survive, and survivors tended to be younger, with most falling below 40 years of age. The diagonal distributions confirmed visible separation between survivors and non-survivors across age, fare and family size variables.

4.4 Statistical Analysis

Descriptive statistics revealed that only 38.38% of passengers survived, indicating that the majority perished in the disaster. The youngest passenger was approximately 0.42 years old, confirming infants were aboard. The wide gap between the mean fare of 32 and the maximum fare of 512 reflects a highly unequal fare distribution. Frequency distribution showed 549 deaths compared to 342 survivors. Third class dominated with 491 passengers, males outnumbered females 577 to 314, and the majority boarded from Southampton port.

The strongest positive correlation was found between SibSp and Parch at 0.41, indicating family groups tended to travel together. The strongest negative correlation was between Pclass and Fare at -0.55. Three important statistical findings emerged: first, the overall survival rate of 38.38% confirms the majority did not survive; second, first class passengers paid an average fare of 84.15 compared to 13.68 for third class, highlighting extreme economic disparity; third, 74% of females survived compared to only 19% of males, strongly confirming that the women and children first evacuation policy was enforced.

4.5 Machine Learning

A logistic regression classifier was developed to predict passenger survival. Seven predictor variables were selected: Pclass, Age, SibSp, Parch, Fare, Sex (encoded), and Embarked (encoded). Categorical variables were transformed using label encoding. The dataset was split 80/20 into training (712 samples) and testing (179 samples) sets with a random state of 42 for reproducibility.

The model achieved an accuracy of 81.01%. The confusion matrix revealed 90 true negatives, 55 true positives, 15 false positives, and 19 false negatives. The model demonstrated stronger recall for non-survivors at 86% compared to survivors at 74%, suggesting a slight bias towards predicting death over survival, likely due to class imbalance in the dataset.

5. Discussion

The analysis of the Titanic dataset yielded several compelling findings that align with historical accounts of the disaster. Perhaps the most striking observation was that despite females being fewer in number, 74% of them survived compared to only 19% of males, strongly confirming that the women and children first evacuation policy was actively enforced. This human decision, rather than random chance, significantly shaped survival outcomes.

A significant fare disparity was observed, with first class passengers paying an average of 84.15 compared to 13.68 for third class passengers. This economic divide extended beyond mere comfort — it appears to have directly influenced survival probability, as wealthier passengers in lower class numbers had better access to lifeboats and rescue resources. The majority of passengers fell within the 20 to 40 age range, with first class passengers recording the highest median age of 35, reflecting the demographic of established, wealthier travellers.

A key limitation of this study is the heavy influence of the women and children first evacuation policy, which skewed survival outcomes in favour of females regardless of other variables. This social rule, rather than purely statistical factors, played a significant role in determining who survived, making it difficult to draw purely data-driven conclusions. Additionally, the high proportion of missing cabin data meant that cabin location, which likely influenced evacuation speed and survival, could not be factored into the analysis.

6. Conclusion

This project successfully demonstrated the application of Python-based data science techniques to a real-world dataset. The logistic regression model achieved an accuracy of 81.01%, confirming that survival outcomes can be predicted with reasonable confidence using demographic and ticket variables. The analysis consistently revealed that gender, passenger class, and fare were the strongest determinants of survival on the Titanic.

Future analysis should investigate whether social connections existed between passengers across different classes, particularly examining whether third class passengers with relatives or acquaintances in first class had higher survival chances. Incorporating passenger name matching and ticket grouping techniques could uncover hidden social connections within the data, potentially revealing that survival was not purely determined by class and fare but also by personal relationships across class boundaries.

7. References

- Kaggle. (2012). Titanic: Machine Learning from Disaster. Retrieved from <https://www.kaggle.com/competitions/titanic/data>
- McKinney, W. (2022). Python for Data Analysis (3rd ed.). O'Reilly Media.
- Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.
- Waskom, M. (2021). Seaborn: Statistical Data Visualization. *Journal of Open Source Software*, 6(60), 3021.

8. Appendix — Python Code

The complete Python code is available in the Jupyter Notebook (titanic_analysis.ipynb) hosted on the GitHub repository. The repository contains the notebook, the dataset (train.csv), this report in PDF format, and a README.md describing the project structure and how to run the analysis.

Key code snippets are summarised below for reference:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
from sklearn.preprocessing import LabelEncoder

df = pd.read_csv('train.csv')
df['Age'] = df['Age'].fillna(df['Age'].median())
df = df.drop(columns=['Cabin'])
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

le = LabelEncoder()
df['Sex_encoded'] = le.fit_transform(df['Sex'])
df['Embarked_encoded'] = le.fit_transform(df['Embarked'])
X =
df[['Pclass', 'Age', 'SibSp', 'Parch', 'Fare', 'Sex_encoded', 'Embarked_encoded']
]
y = df['Survived']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print(accuracy_score(y_test, y_pred))
```